Field-Agnostic Four-Level Human-Supervised Decision-Support System

Sponsors: Facility for Rare Isotope Beams (FRIB) at Michigan State University
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Project Description

Various systems in particle accelerator facilities generate complex, high-volume data during trip events and routine operations. Operators and experts must rapidly analyze raw waveforms, cross-reference operating procedures, and record institutional knowledge under tight operational timelines.

This capstone project aims to design and implement a functional end-to-end proof-of-concept (POC) for a field-agnostic Four-Level Human-Supervised Decision-Support System. The architecture is explicitly non-autonomous, serving strictly to augment human operators by structuring raw evidence, running diagnostics, retrieving procedural context, and drafting actionable event summaries for human approval. It will be built using standardized data contracts (e.g., canonical JSON schemas / APIs) capable of accepting time-series signals, discrete state logs, and procedural documents from any engineering domain.

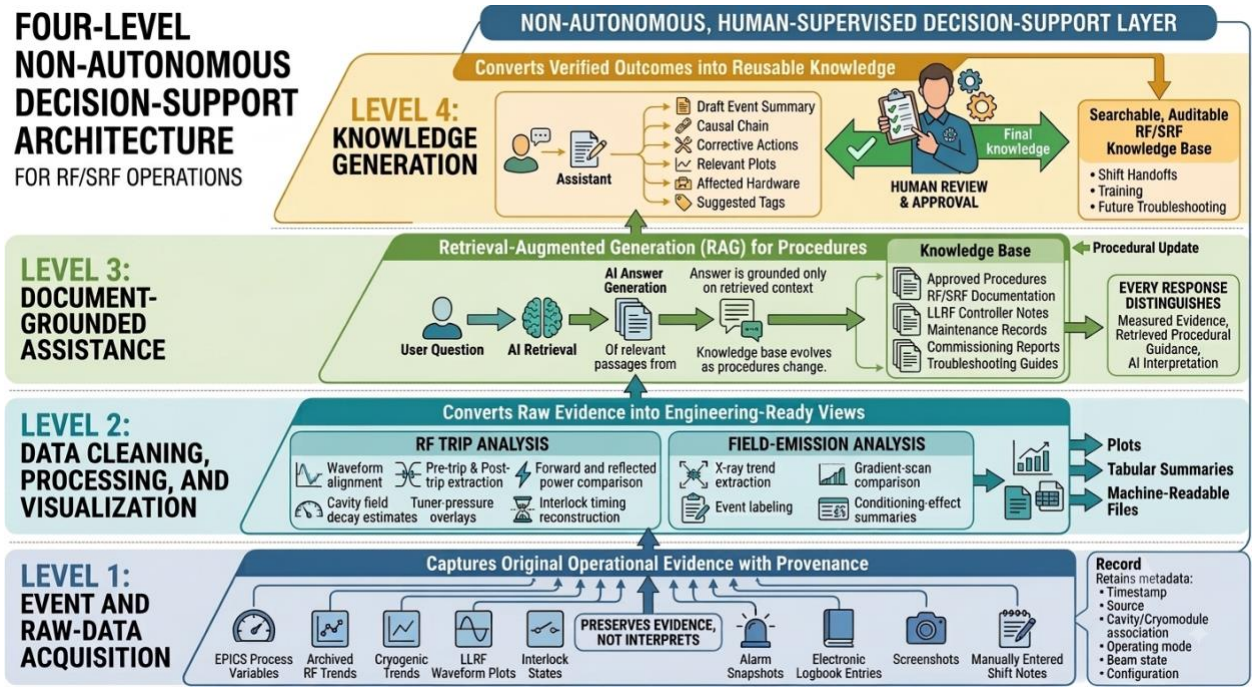


Figure 1. Example of Four level decision support architecture for RF/SRF operations

Because live operational data is unavailable, the team will design and build a Synthetic Engineering Data Generator to simulate multi-domain operational events, interlocks, and

telemetry. The resulting platform will act as a modular engine ready to accept real-world data pipelines upon deployment.

The system spans four integrated layers:

- Level 1 (Data Acquisition & Metadata Preservation): Ingestion & Provenance.
- Level 2 (Data Processing & Visualization): Analytics & Plotting.
- Level 3 (Document-Grounded Assistance): A Retrieval-Augmented Generation (RAG) engine grounded exclusively in official facility procedures and manuals.
- Level 4 (Knowledge Generation & Human Decision Gate): Automated draft summary generation with an explicit human review gate for entry into a searchable knowledge base.

Major Work Items

Phase 1: Data Contracts & Synthetic Generator Framework

Phase 2: Plug-and-Play Processing Subsystems

Phase 3: Knowledge Synthesis, UI, & Decision Gates

Phase 4: Validation, Testing, & Final Presentation

Expected Deliverables

- Synthetic Engineering Data Generator: A standalone, scriptable tool capable of generating realistic time-series, event logs, and metadata for multiple engineering fields (RF, Mechanical, PLC, etc.).
- Standardized Data Interface Specification: Comprehensive API and schema documentation (OpenAPI / JSON Schema) defining the plug-and-play ingestion contracts.
- Field-Agnostic Decision-Support Core: Fully modular source code repository housing:
    - Level 1/2 Engine: Generic data cleaning, alignment, FFT, and plotting services.
    - Level 3 Engine: Domain-flexible RAG framework.
    - Level 4 Engine: Knowledge generation and hypothesis ranking modules.
- Interactive Dashboard & Human Review Interface: Web platform providing visual data exploration, RAG query interfaces, and the explicit "Approve / Correct / Escalate" decision gate.
- Multi-Domain Validation Suite & Final Report: Test results demonstrating the system successfully handling data across at least 3 distinct engineering domains, accompanied by deployment documentation (Docker setup).

Software/Package Requirements

100% Free / Local Stack (Streamlit, ChromaDB, SentenceTransformers, Ollama or Gemini Free Tier, and SQLite). This eliminates API billing friction, requires no budget approval and allows the entire decision-support pipeline to be built, tested, and demonstrated locally via Docker Compose.

Project Duration

End of August to Early December